Experiment no - 4

```
import  pandas as pd
import numpy as np
```

Reading data using read_csv() function

```
df=pd.read_csv('/content/diamonds1.csv')
```

Displaying the DataFrame

```
df
```

Show hidden output

Next steps: Generate code with df | New interactive sheet

Finding the data types of variables in the DataFrame

```
df.dtypes
```

Show hidden output

Importing libraries essential for data visualization

```
#MATPLOTLIB
import matplotlib.pyplot as plt
%matplotlib inline
#SEABORN
import seaborn as sns
```

.

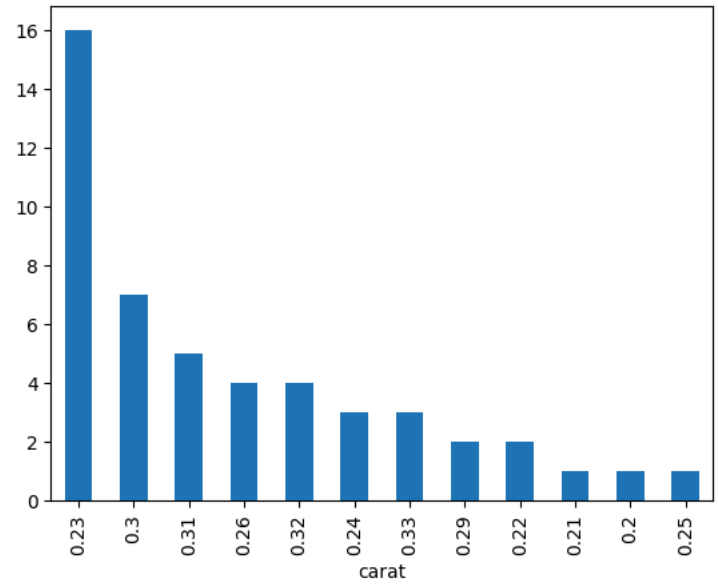
Plots for categorical variables' analysis

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BAR PLOT

```
df['carat'].value_counts().plot.bar()
```

```
<Axes: xlabel='carat'>
```

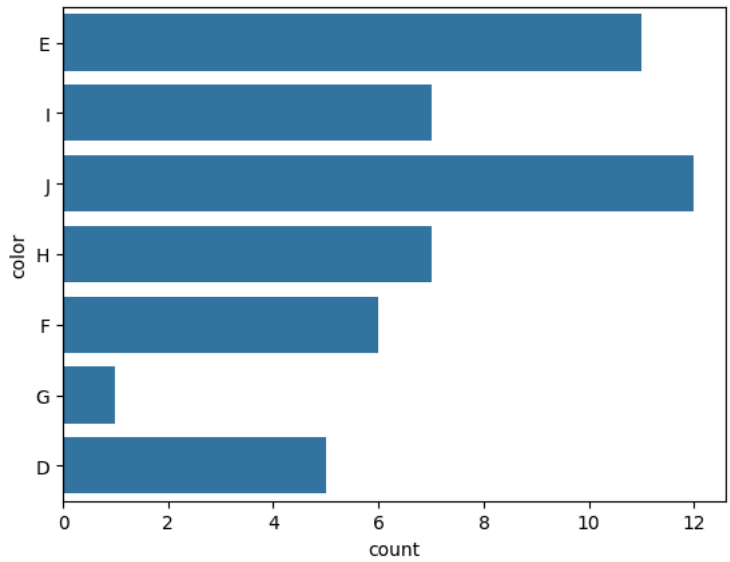


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```
sns.countplot(df['color'])
```

```
<Axes: xlabel='count', ylabel='color'>
```

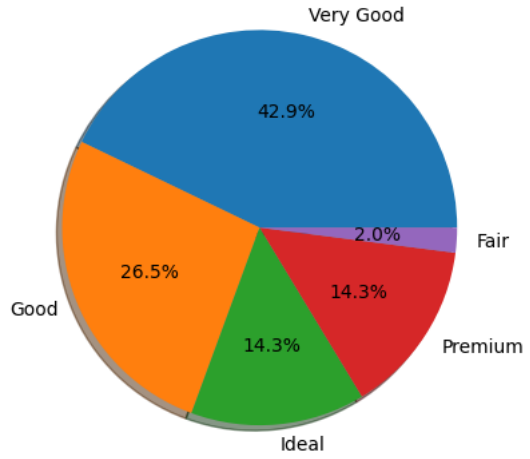


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```
plt.pie(df['cut'].value_counts(), labels=df['cut'].value_counts().index, autopct='%1.1f%%', shadow=True)
```

```
([<matplotlib.patches.Wedge at 0x7ed7374dfa70>,
  <matplotlib.patches.Wedge at 0x7ed73767a390>,
  <matplotlib.patches.Wedge at 0x7ed737282180>,
  <matplotlib.patches.Wedge at 0x7ed739d7a2d0>,
  <matplotlib.patches.Wedge at 0x7ed73a2afa40>],
 [Text(0.2447729506452284, 1.072420720907811, 'Very Good'),
  Text(-1.0196083041990471, -0.4127940237071554, 'Good'),
  Text(0.10562553361244671, -1.0949170044569982, 'Ideal'),
  Text(0.9218972471750501, -0.6000878816065734, 'Premium'),
  Text(1.0977399473085168, -0.070477003931036, 'Fair')],
 [Text(0.13351251853376092, 0.5849567568588059, '42.9%'),
  Text(-0.5561499841085711, -0.2251603765675393, '26.5%'),
  Text(0.05761392742497093, -0.5972274569765444, '14.3%'),
  Text(0.5028530439136636, -0.3273206626944946, '14.3%'),
  Text(0.5987672439864636, -0.03844200214420145, '2.0%')])
```

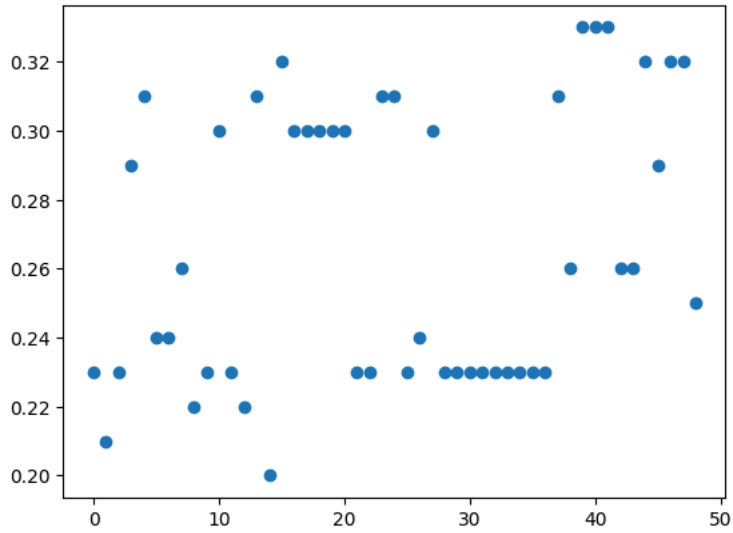


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Plots for continuous variables' analysis

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```
plt.scatter(df.index,df['carat'])
plt.show()
```

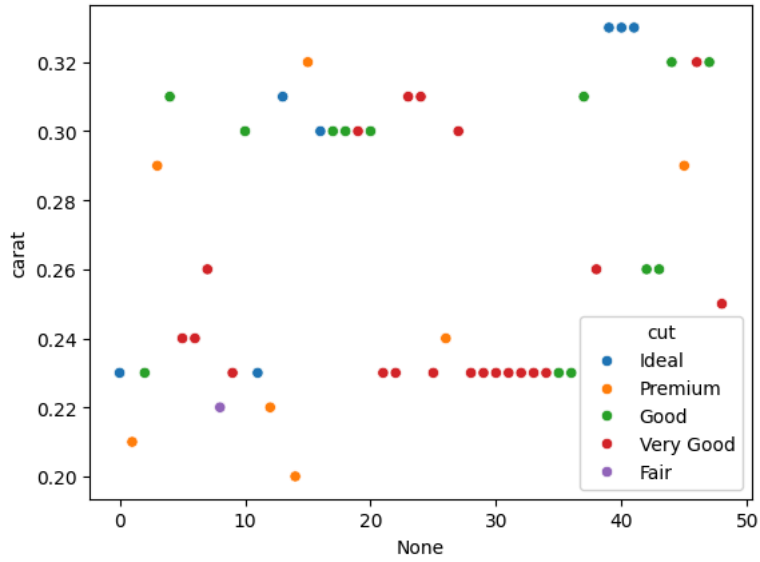


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```
sns.scatterplot(x=df.index,y=df['carat'],hue=df['cut'])
```

```
<Axes: xlabel='None', ylabel='carat'>
```



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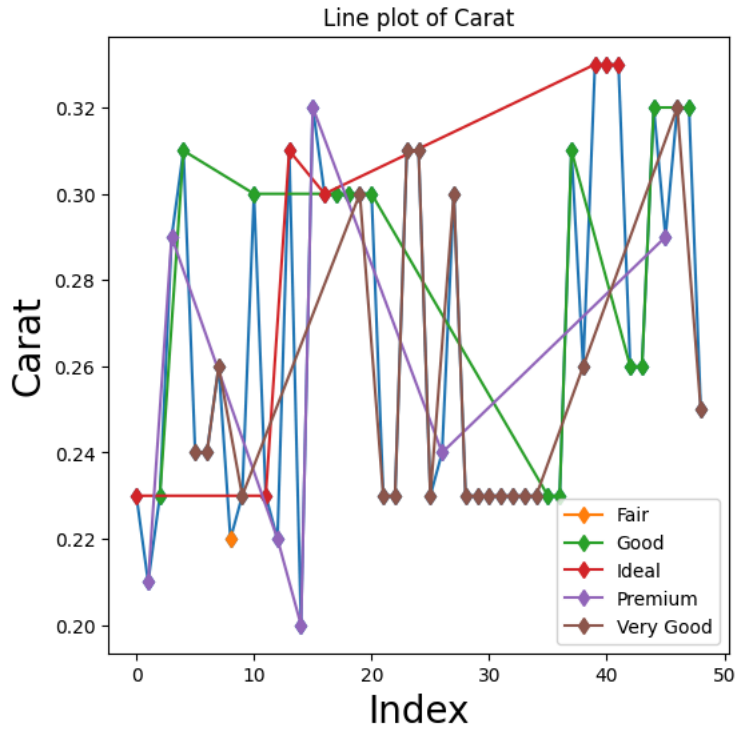
.

```
plt.figure(figsize=(6,6))
plt.title('Line plot of Carat')
plt.xlabel('Index', fontsize=20)
plt.ylabel('Carat', fontsize=20)

plt.plot(df.index, df['carat'], markevery=1, marker='d')

for name, group in df.groupby('cut'):
    plt.plot(group.index, group['carat'], label=name, markevery=1, marker='d')

plt.legend()
plt.show()
```

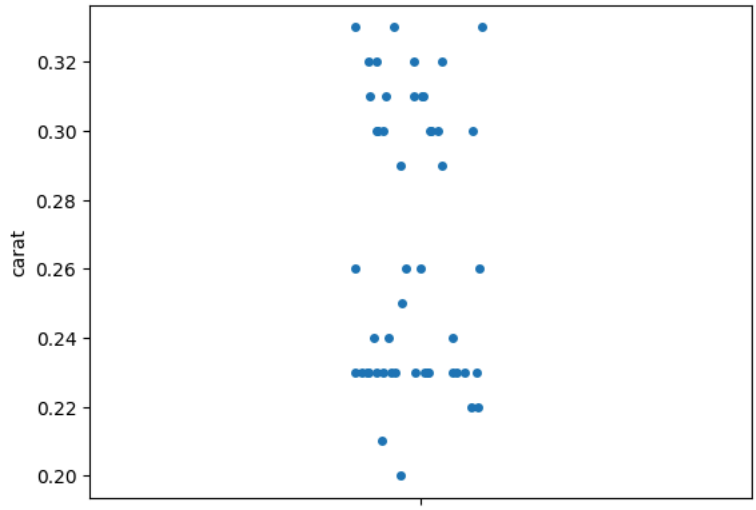


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```
sns.stripplot(y=df['carat'])
```

```
<Axes: ylabel='carat'>
```

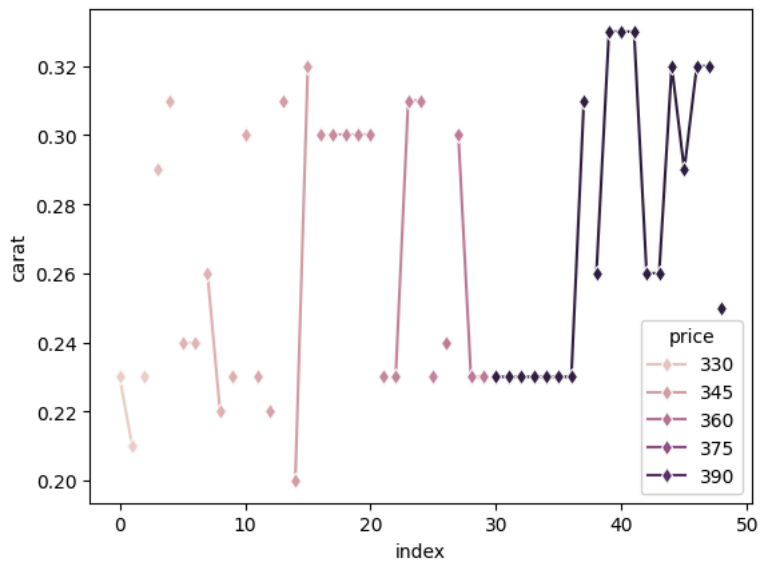


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```
fig=sns.lineplot(x=df.index,y=df['carat'],markevery=1,marker='d',data=df,hue=df['price'])
fig.set(xlabel='index')
```

```
[Text(0.5, 0, 'index')]
```



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```
sns.distplot(df['carat'],rug=True,hist=False)
```

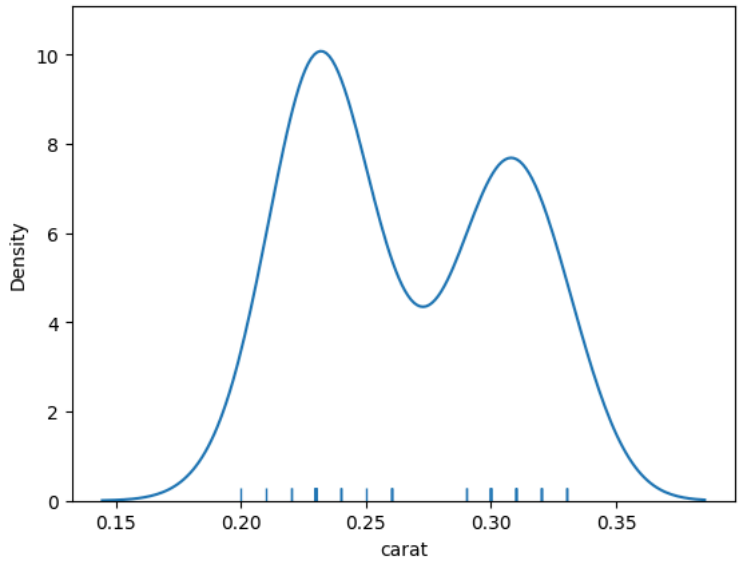
```
/tmp/ipython-input-3366706619.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `kdeplot` (an axes-level function for kernel density plots).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(df['carat'],rug=True,hist=False)
<Axes: xlabel='carat', ylabel='Density'>
```



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```
plt.boxplot(df['table'])
```

```
{'whiskers': [<matplotlib.lines.Line2D at 0x7ed7367902c0>,
  <matplotlib.lines.Line2D at 0x7ed736790830>],
 'caps': [<matplotlib.lines.Line2D at 0x7ed736790b30>,
  <matplotlib.lines.Line2D at 0x7ed736790e30>],
 'boxes': [<matplotlib.lines.Line2D at 0x7ed736790440>],
 'medians': [<matplotlib.lines.Line2D at 0x7ed736791130>],
 'fliers': [<matplotlib.lines.Line2D at 0x7ed7367913d0>],
 'means': []}
```

